AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 36 · STUDENT LESSON

Neural network experiment defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Show the calculation. Name the comparison. Preserve the limit.

Defend a neural experiment using its full evidence chain and explain what remains unmeasured.

Keep one habit: Inventory → stage claims → complete final rows → challenge responses → separate neuron → executable chain → independent defense.

MAIN LESSON · SECTION 1

Defend the evidence chain, including its limits

> A strong explanation makes the calculations and decisions checkable.

Learning goals

I can defend a network experiment using traceable **evidence**.

I can separate selection, final evaluation and revision claims.

I can answer a challenge with a calculation and a bounded conclusion.

Retrieve capstone evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone **limitation**s and revision. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Can a revised model reuse inspected final rows as new independent **evidence**?

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

Explain neural network experiment **defense** with a traceable toy-network case and a bounded conclusion.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Defend a network experiment using traceable **evidence**

Separate selection, final evaluation and revision claims

Answer a challenge with a calculation and a bounded conclusion

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Assemble the complete question-to-result record

> Keep validation selection separate from the frozen final comparison.

Evidence inventory and original capstone result

A **defense** is not a sales pitch. Assemble the original question and criterion, data roles, baseline, model equation, settings, raw checkpoints, validation selection, frozen final **prediction**s, **limitation**s and revision history.

Someone reading the record should be able to recompute a reported MSE and see which decisions each dataset informed.

For this capstone, two rates were compared under matched conditions. Rate 0.2 selected a = 2 with validation MSE 0 versus 0.78125 for rate 0.1.

The frozen candidate then achieved final MSE 5 versus baseline 20.25 on four constructed cases. These are different comparisons and must not be collapsed into “the network was 100% accurate”.

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Attach a source and a limit to each claim

> A reader should be able to recompute the MSE rather than trust a slide.

Stage-by-stage evidence and permitted conclusions

Stage	Evidence	Supported statement
Validation selection	Two positive rows; MSE 0 versus 0.78125	Selected rate 0.2 by written rule
Frozen final comparison	Four rows; MSE 5 versus 20.25	Lower MSE than initial-weight baseline on these rows
Condition check	Two negative rows; candidate MSE 10	Nonnegative output fails to represent negative targets
Proposed revision	Signed output on revealed cases	A possible representational repair, not new final evidence

Complete original frozen final rows

Final ID · x · Target · Baseline prediction · Candidate prediction · Baseline squared error · Candidate squared error

Final ID: F1

x: 5

Target: 10

Baseline prediction: 5

Candidate prediction: 10

Baseline squared error: 25

Candidate squared error: 0

Final ID: F2

x: 6

Target: 12

Baseline prediction: 6

Candidate prediction: 12

Baseline squared error: 36

Candidate squared error: 0

Final ID: F3

x: -1

Target: -2

Baseline prediction: 0

Candidate prediction: 0

Baseline squared error: 4

Candidate squared error: 4

Final ID: F4

x: -2

Target: -4

Baseline prediction: 0

Candidate prediction: 0

Baseline squared error: 16

Candidate squared error: 16

Final ID: Four-row MSE

x: Not classification accuracy

Target: Same cases

Baseline prediction: 20.25

Candidate prediction: 5

Baseline squared error: Negative candidate subset

Candidate squared error: 10 on two rows

Follow the evidence-role lanes

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Question and settings	Replay calculations	Procedure can be checked
Frozen final predictions	Compare same four rows	Bounded baseline improvement
Known failures and changes	Preserve old results	Revision still needs evaluation

Every claim has an evidence source and a limit. A complete record is not a substitute for broader data.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Answer a challenge with evidence and a bounded statement

> Keep an absent comparison or measurement explicitly absent.

Current task and response

- State the validation selection and frozen final comparison using their correct metrics and scopes.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the changed neuron

- Change: Change the second **input** from 1 to 2.

▼ **Check after predicting**

Prediction: z becomes 1-2+0.2=-0.8 and ReLU returns 0.

Reason: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Separate a neuron demonstration from a system evaluation

> A correct numerical trace does not supply a population-performance claim.

Main negative-target challenge and answer

Challenge: “Your validation error was zero, so why did the final test fail on negative targets?” Response: validation contained only positive **input**s; the **output** max(0,a*x) cannot be negative.

The final negative cases expose a coverage and architecture **limitation**. The overall criterion still passed, but that does not remove the failure.

Stronger-baseline and speed challenges

Challenge: “Is a neural network better than linear regression?” Response: the supplied baseline was a fixed initial weight, not a fitted line. The **evidence** establishes improvement relative to that baseline only.

A fitted signed line is a relevant additional comparison and needs a documented, fair evaluation. Challenge: “Was the candidate faster?” Response: no timing **evidence** was collected, so speed is unmeasured.

Separate pictured neuron inputs and scope

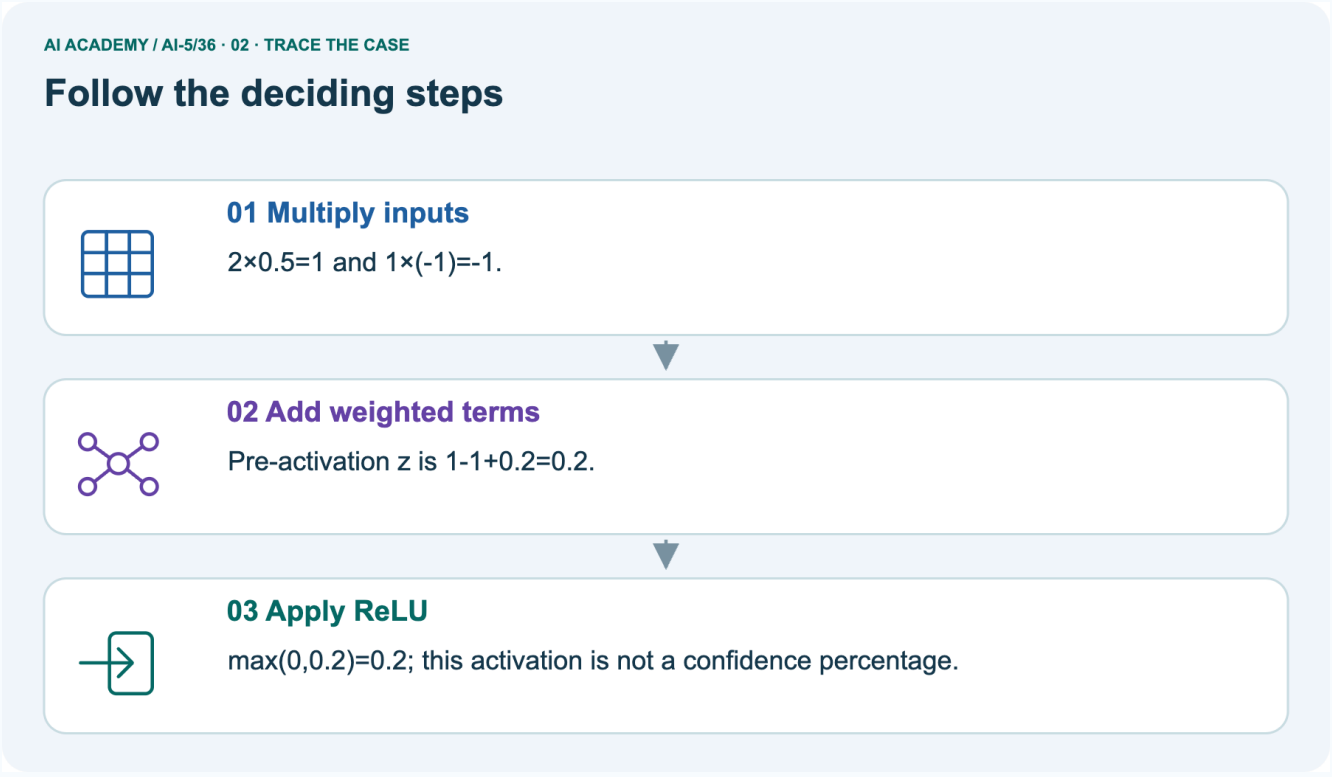
Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

Inputs are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

The picture is a separate arithmetic demonstration, not the scalar capstone's evaluation. Inputs [2, 1], weights [.5, –1], bias .2 give weighted terms 1 and –1, preactivation .2 and ReLU output .2. Changing only the second input to 2 gives z = –.8 and output 0. There is no supplied class-reference dataset, probability conversion or accuracy denominator for this neuron.

Its output .2 cannot be called 20% confidence or 20% accuracy.

Trace multiplication, bias and activation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Multiply inputs	2×0.5=1 and 1×(-1)=-1.
2. Add weighted terms	Pre-activation z is 1-1+0.2=0.2.
3. Apply ReLU	max(0,0.2)=0.2; this activation is not a confidence percentage.

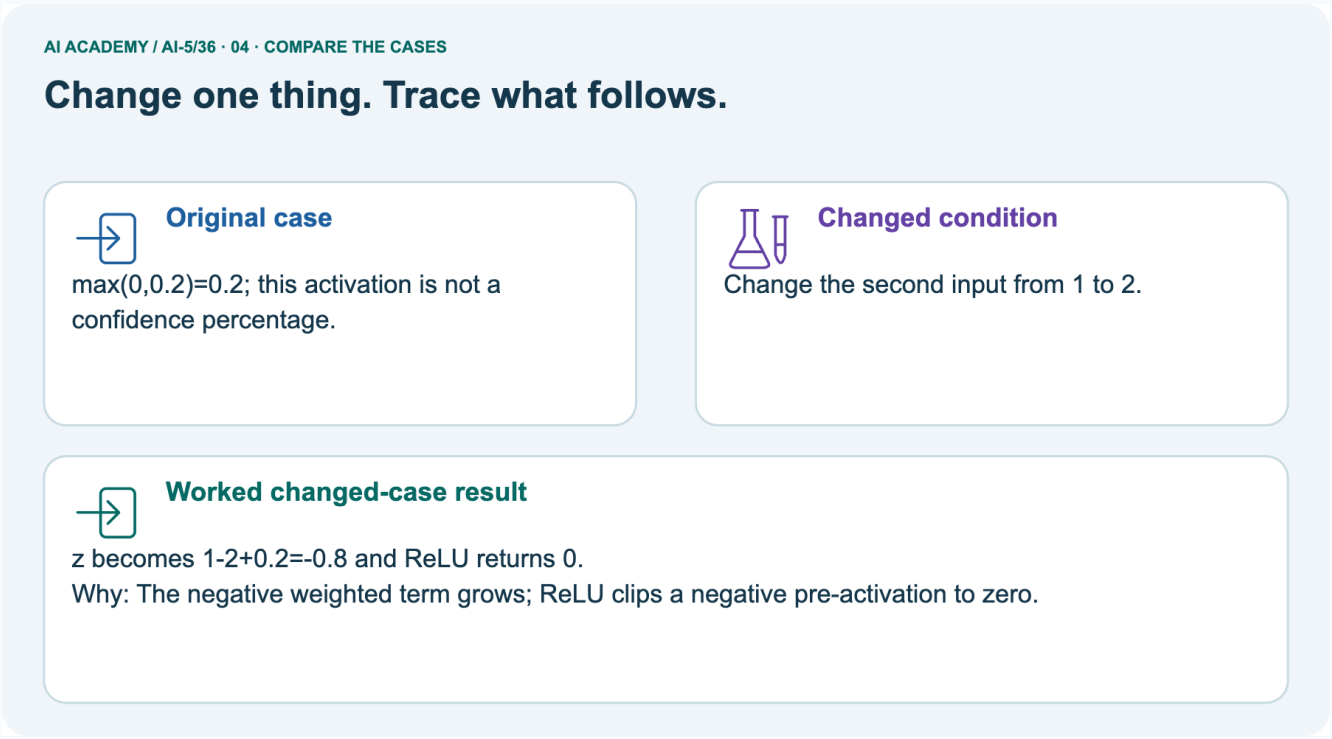
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

Predict the changed second input

- Change one thing: Change the second **input** from 1 to 2. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain unsupported percentage and generalisation claims

- This one-neuron arithmetic does not establish how a trained image classifier generalises or why its full network makes a decision.
- Read the labelled picture: Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Neural network experiment defense

The facts we will use

AI ACADEMY / AI-5/36

One toy ReLU neuron: not a probability

Input 2 x weight 0.5 = 1

Input 1 x weight -1 = -1

Add the products and bias 0.2: total = 0.2

ReLU: use max(0, total) -> output 0.2

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/36

Neural network experiment defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Multiply inputs

2 x 0.5=1 and 1 x (-1)=-1.





2. Add weighted terms

Pre-activation z is 1-1+0.2=0.2.





3. Apply ReLU

max(0,0.2)=0.2; this activation is not a confidence percentage.

CHANGE ONE THING

Change the second input from 1 to 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Keep relative error reduction attached to its metric

> A 75% MSE reduction is not a classification-accuracy score.

Complete separate supported comparison and task

- Supported case: another experiment reports baseline MSE 8 and candidate MSE 2 on four untouched final rows after freezing the candidate. The reduction is 6 MSE units, or 75% of the baseline MSE.

  That percentage is a relative reduction in MSE, not 75% classification accuracy. It cannot establish deployment performance.

> **Guided: calculate the absolute and relative MSE reduction, then write a sentence preserving the metric and four-row scope.**

Identify missing fitted-line and energy evidence

- A presenter claims the experiment proves neural networks beat fitted lines and use less energy. What **evidence** is missing?

> **State one repair and the checks you would rerun.**

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise an evidence inventory and an honest defense

> Every claim needs a metric, cases and a comparison.

Current inventory task and supported reduction

- Name eight items needed to check the capstone from question through final result.

Guided: calculate the absolute and relative MSE reduction, then write a sentence preserving the metric and four-row scope.

Explain your prediction to a partner before comparing results.

Supported W2 record

Use workbook W2 for the connected example “Neural network experiment **defense**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual demonstration-versus-evaluation check

- Use the worked case for Neural network experiment **defense**. Proposed explanation: “A polished demonstration is an evaluation”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured neuron response

- Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

> **My explanation using a pictured step and the given facts**

Reserve independent C1/C2 record

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Defend the result

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Replay the complete chain and audit each claim

> Check numerical agreement and preserve the limits on what that agreement establishes.

Current validation-versus-final task

State the validation selection and frozen final comparison using their correct metrics and scopes.

Complete executable evidence-chain replay

```
"""Replay the supplied capstone evidence chain; no population claim."""
train=[(1,2),(2,4)]
validation=[(3,6),(4,8)]
final=[(5,10),(6,12),(-1,-2),(-2,-4)]
def predict(weight,x):return max(0,weight*x)
def mse(weight,rows):return sum((predict(weight,x)-y)**2 for x,y in rows)/len(rows)
records=[]
for rate in [.1,.2]:
    weight=1.0
    history=[]
    for update in [1,2]:
        assert all(weight*x>0 for x,_ in train)
        gradient=sum(2*(weight*x-y)*x for x,y in train)/len(train)
        weight-=rate*gradient
        history.append((update,gradient,weight))
    records.append({'rate':rate,'history':history,'weight':weight,
                    'validation_mse':mse(weight,validation)})
selected=min(records,key=lambda r:(r['validation_mse'],r['rate']))
```

```
frozen_weight=selected['weight']
assert frozen_weight==2 and selected['rate']==.2
# No updates or choices below this point use the final targets.
final_ledger=[(x,y,predict(1,x),predict(frozen_weight,x),
               (predict(1,x)-y)**2,(predict(frozen_weight,x)-y)**2)
              for x,y in final]
baseline_mse=mse(1,final)
candidate_mse=mse(frozen_weight,final)
negative_mse=mse(frozen_weight,[(x,y) for x,y in final if x<0])
assert (baseline_mse,candidate_mse,negative_mse)==(20.25,5,10)
print('Validation selection records:',records)
print('Frozen final: x,target,baseline,candidate,baseline squared error,candidate squared
error',final_ledger)
print('Final MSE baseline/candidate:',baseline_mse,candidate_mse)
print('Negative-input MSE, 2 rows:',negative_mse)
print('Scope: two positive validation rows; four constructed final rows.')
print('Unmeasured: timing, energy, fitted-line comparison, independent revised-model
performance.')
# Separate supported comparison: not the main scores.
absolute=8-2;relative=absolute/8
assert (absolute,relative)==(6,.75)
print('Separate supported reduction:',absolute,'MSE units;',100*relative,'% relative MSE
reduction')
# Pictured neuron: an activation, not a probability or an accuracy percentage.
def neuron(second):
    z=2*.5+second*(-1)+.2
    return z,max(0,z)
assert abs(neuron(1)[0]-.2)<1e-12 and neuron(2)[1]==0
print('Picture original z/output:',neuron(1),'changed:',neuron(2))
```

Expected checks and exposure-aware conclusions

1. Predict the validation choice and all final squared errors before running or tracing. This supplied replay reconstructs both training histories, selects on validation, freezes the chosen weight and then computes final scores without using final targets to update or choose parameters. Preserve the original data exposure history: a public teaching replay does not establish that a learner's own final cases were unseen.

   Keep the separate 8→2 MSE example and the separate neuron demonstration labelled. No timing, energy, fitted-line comparison or independent revised-model result is supplied. Save actual run status and question answers rather than treating a polished presentation as proof.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: defend a fresh C1/C2 evidence record

> Check overall and condition-specific scores, including any worsened row.

Complete independent selection and error inputs

A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The **written rule** selects lower MSE.

After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Defend a new record independently

Attempt this without hints

A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The **written rule** selects lower MSE.

After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep a final defense portfolio with the revision history

> Include the unsuccessful conditions and unmeasured outcomes.

Project task and artifact

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0.

The **written rule** selects lower MSE. After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish by showing what the evidence supports

> End with a useful next comparison and the evaluation it still needs.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Defend a network experiment using traceable **evidence**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Defense	An explanation of methods and claims supported by inspectable evidence
Evidence chain	The connection from inputs and procedure through raw outputs to a reported claim
Limitation	A condition restricting what evidence can establish
Reproduction	Repeating the documented procedure to check reported outputs

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Name eight items needed to check the capstone from question through final result.
- A presenter claims the experiment proves neural networks beat fitted lines and use less energy. What **evidence** is missing?
- Propose a fair next comparison with a fitted signed line. State how to preserve previous **evidence** and reserve new evaluation.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent C1/C2 error record for the first individual attempt.

Current explanation and response

- Use a short presentation: question; method and controls; selection; final result; failures; revision; next question. State the small constructed sample, absent real-world coverage, limited baseline and unmeasured resource use.

  A useful next question is whether a signed-**output** procedure improves on a fitted-line baseline under newly reserved evaluation conditions.

  Passing this capstone means explaining the **evidence** correctly, including an unsuccessful or limited result.

- Propose a fair next comparison with a fitted signed line. State how to preserve previous **evidence** and reserve new evaluation.

> **Exit: correct “our network is universally reliable, 100% accurate, faster than every baseline and repaired on the final test”. Give measured claims and identify the missing evidence.**

> **My limited claim and a useful next check**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Question and settings	Replay calculations	Procedure can be checked
Frozen final predictions	Compare same four rows	Bounded baseline improvement
Known failures and changes	Preserve old results	Revision still needs evaluation

Every claim has an evidence source and a limit. A complete record is not a substitute for broader data.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**
>
> _____
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

By the end, I can	How I will show it
Defend a network experiment using traceable evidence	Explain the deciding step, show a trace or test, and state one limit.
Separate selection, final evaluation and revision claims	Explain the deciding step, show a trace or test, and state one limit.
Answer a challenge with a calculation and a bounded conclusion	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.
- Which labelled step matches this detail? max(0,0.2)=0.2; this activation is not a confidence percentage.
- 1. Add weighted terms
- 2. Apply ReLU
- 3. Multiply **input**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply ReLU max(0,0.2)=0.2; this activation is not a confidence percentage.

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

Changed-case question and optional hint

- Changed-case practice: Change the second **input** from 1 to 2. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. z becomes 1-2+0.2=-0.8 and ReLU returns 0.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: z becomes 1-2+0.2=-0.8 and ReLU returns 0. The negative weighted term grows; ReLU clips a negative pre-activation to zero.

Repair and retain the record

Review Capstone **limitation**s and revision (ai-5/35). Goal: Evaluate a frozen candidate and baseline on final rows. Short practice: Calculate the slice rates and explain the claim revision.

If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?. Return to this week and explain what you changed.

Review Capstone hypothesis and baseline (ai-5/33). Goal: Write a testable capstone hypothesis and a credible baseline. Short practice: Name the intended change, the baseline and the measured outcome.

Why must both versions avoid **training** on the eight reserved labels?. Return to this week and explain what you changed.

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

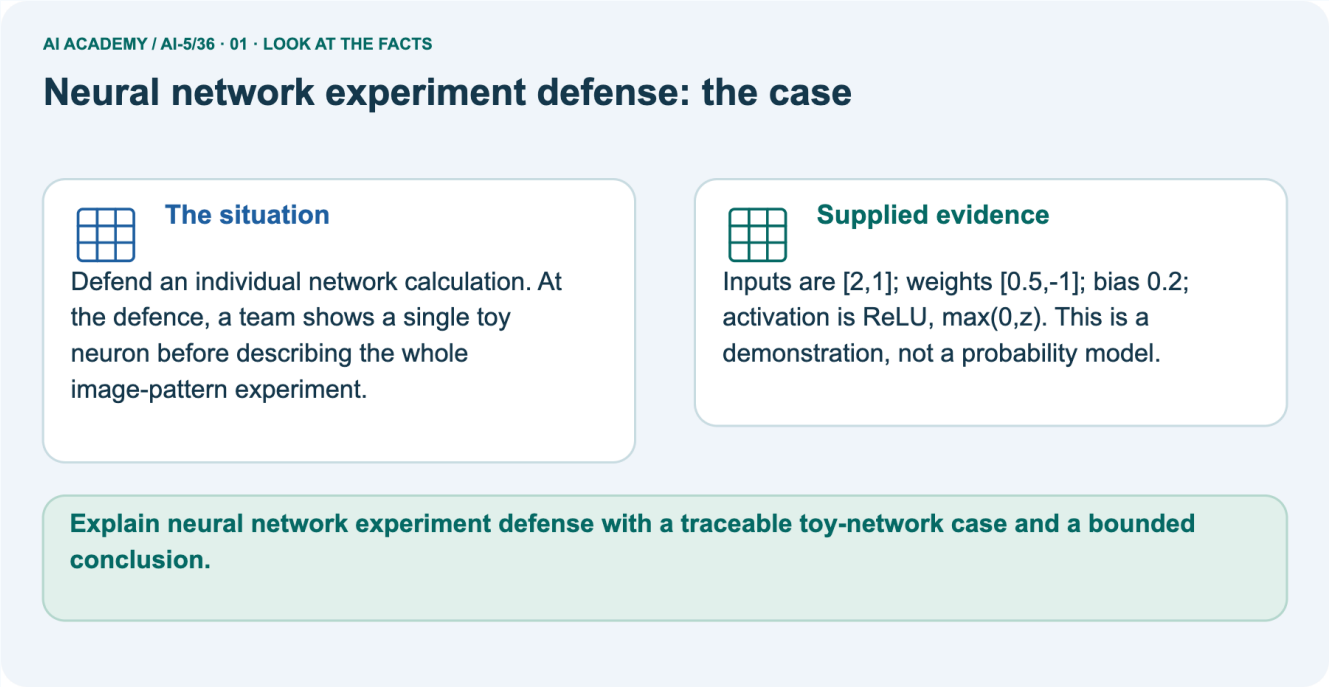
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Neural network experiment defense: the case



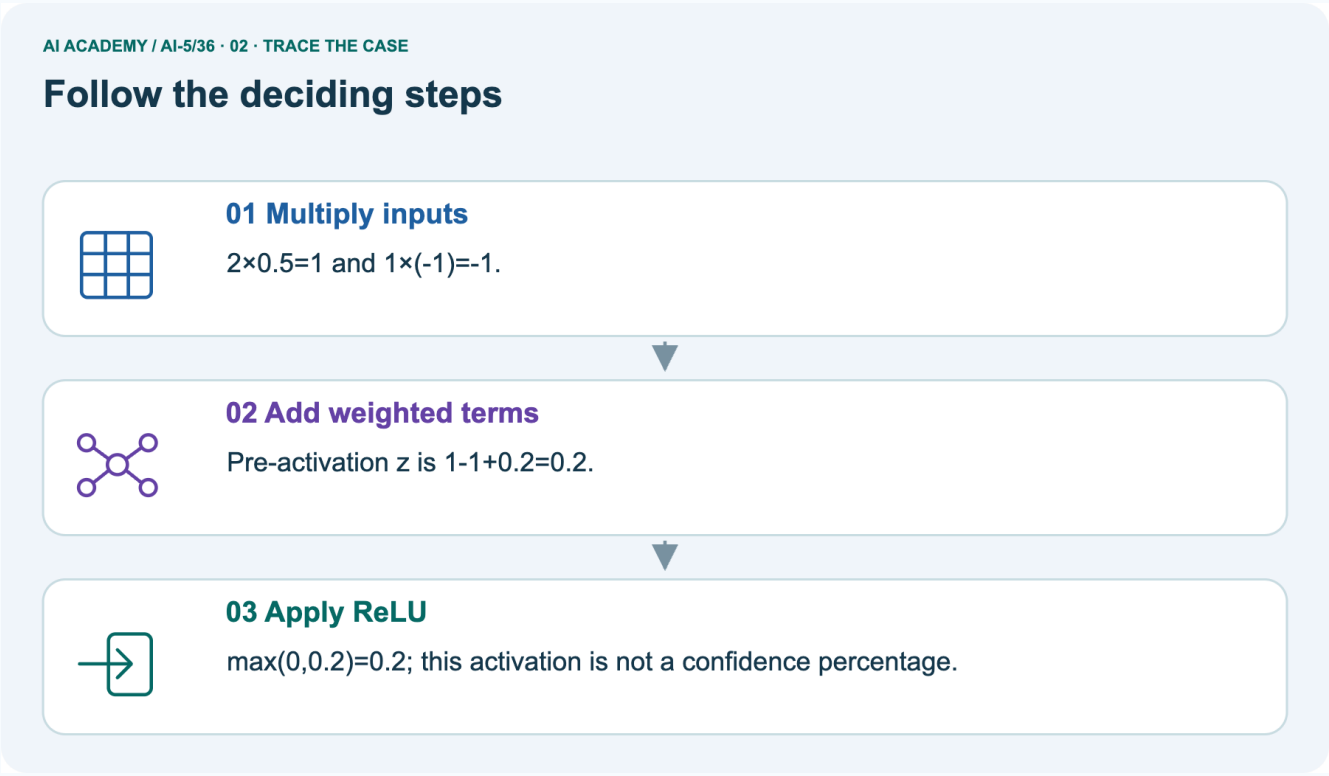
Follow the labelled picture with the matching explanation. Open the vector diagram

Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

Supplied facts: **Input**s are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

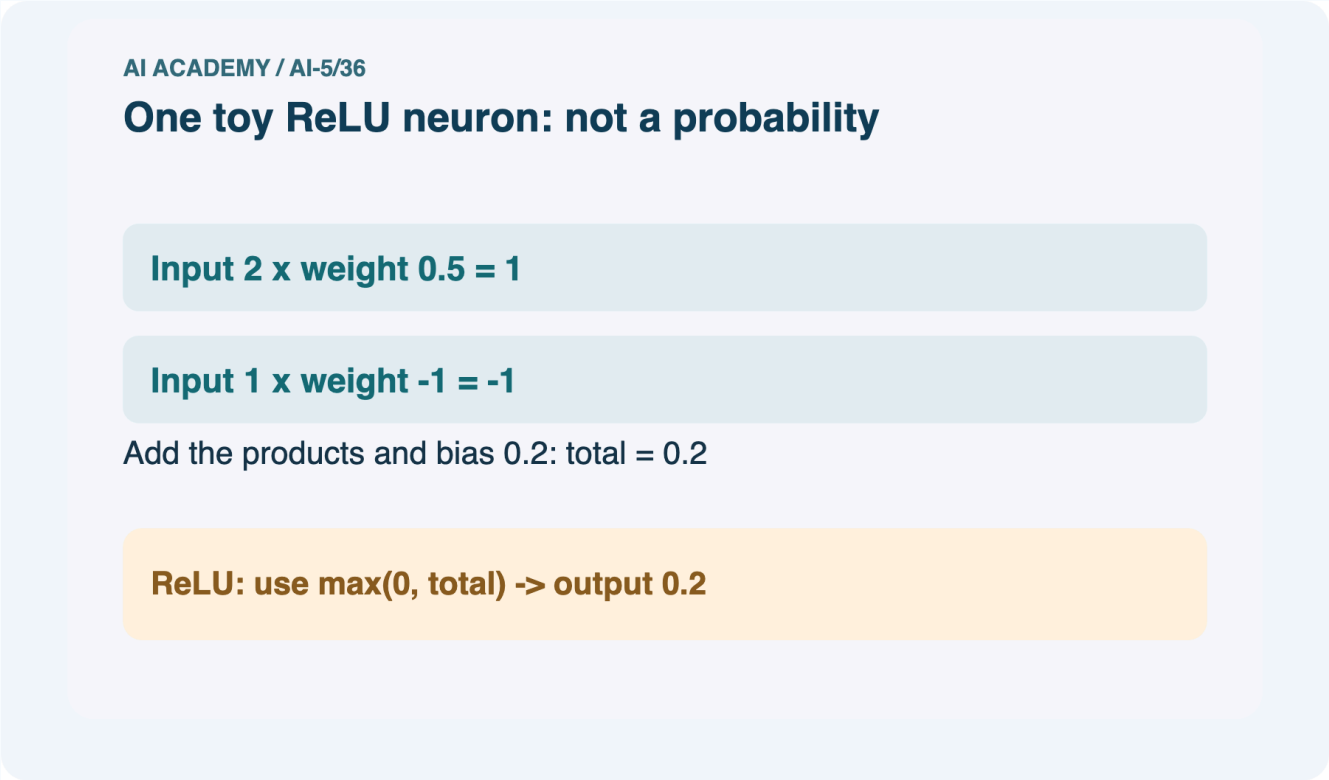


Follow the labelled picture with the matching explanation. Open the vector diagram

Multiply **input**s: 2×0.5=1 and 1×(-1)=-1. → Add weighted terms: Pre-activation z is 1-1+0.2=0.2. → Apply ReLU: max(0,0.2)=0.2; this activation is not a confidence percentage.

Point to each numbered stage and explain which recorded **input** it uses.

03. One toy ReLU neuron: not a probability

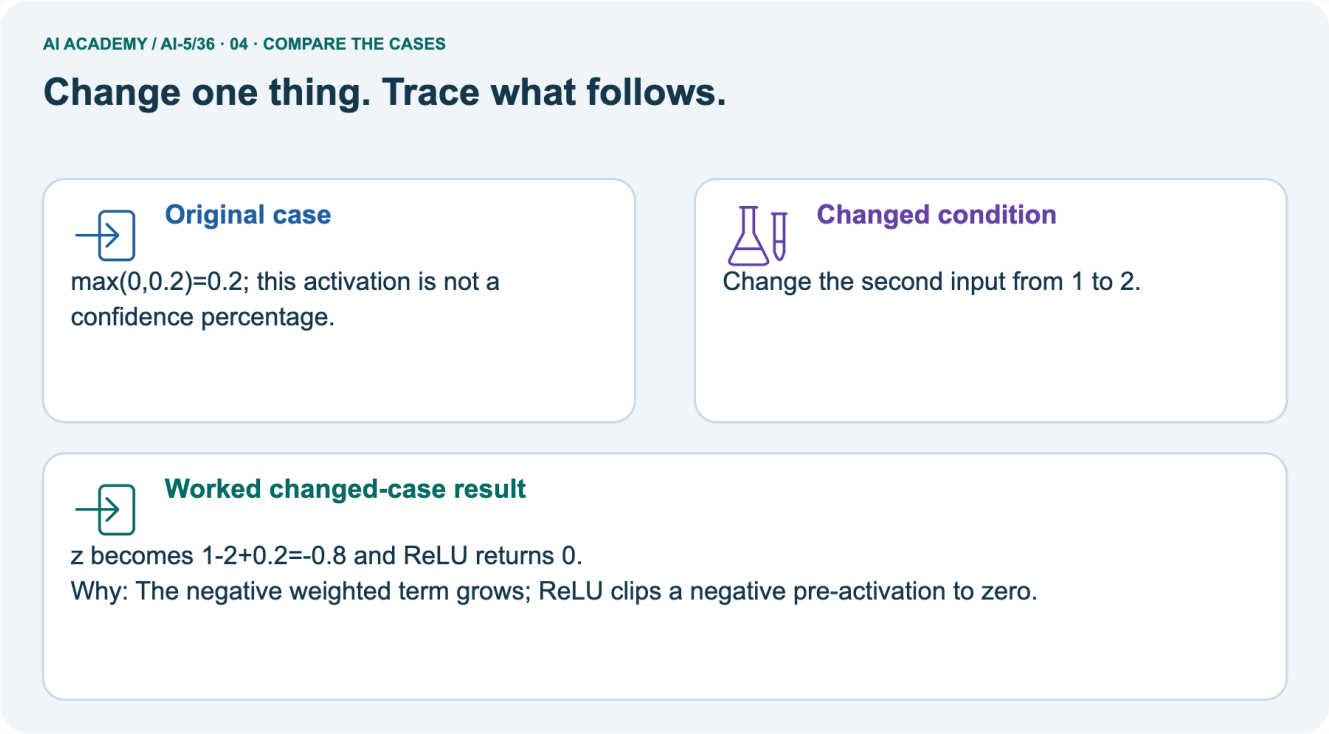


Follow the labelled picture with the matching explanation. Open the vector diagram

One toy ReLU neuron: not a probability. **Input**s: [2, 1]; weights: [0.5, -1]; bias: 0.2; activation: relu; **output**: 0.2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



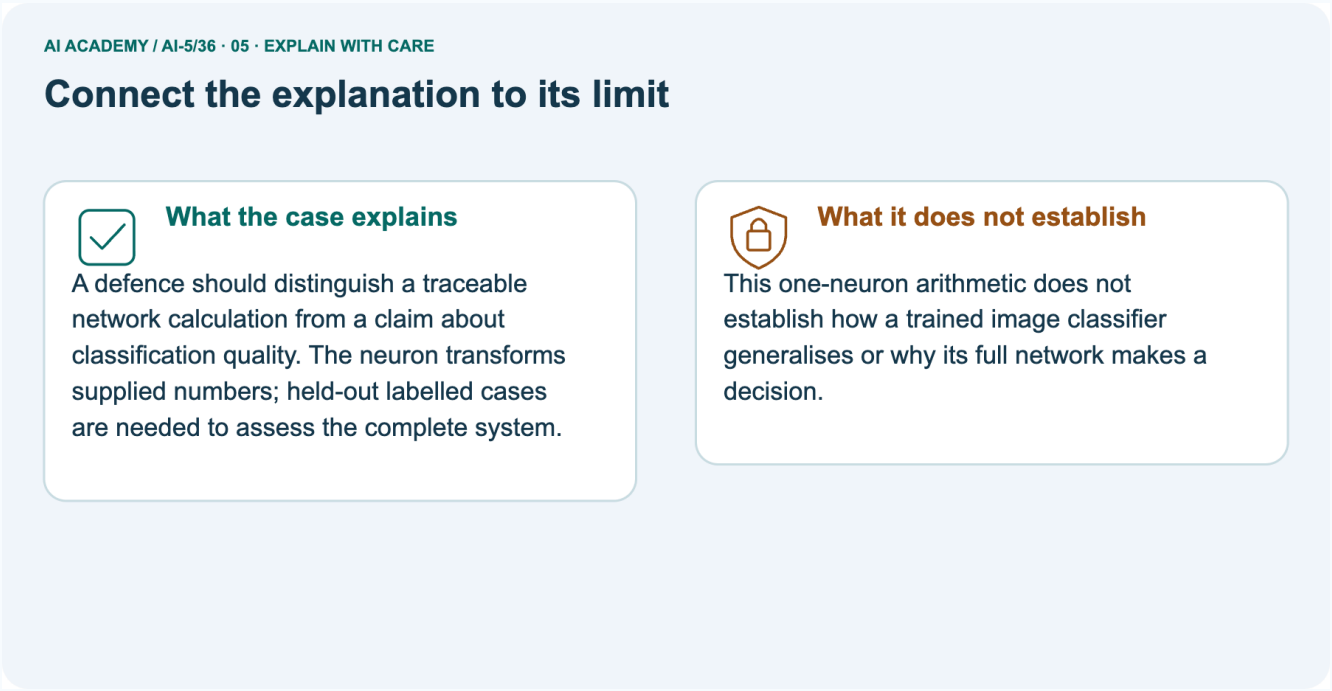
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: max(0,0.2)=0.2; this activation is not a confidence percentage. Changed condition: Change the second **input** from 1 to 2. Expected result: z becomes 1-2+0.2=-0.8 and ReLU returns 0.

Reason: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



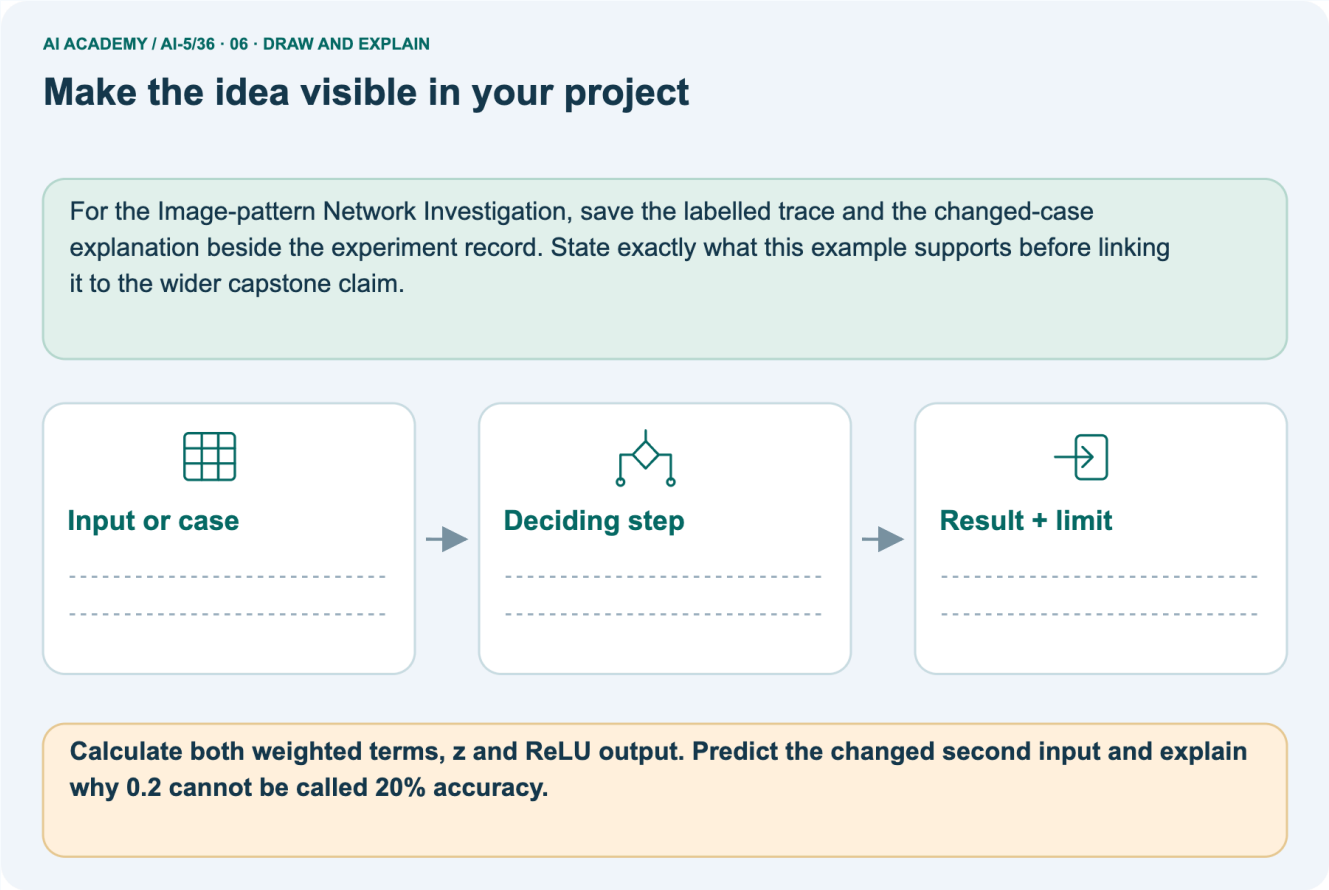
Follow the labelled picture with the matching explanation. Open the vector diagram

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

Limit: This one-neuron arithmetic does not establish how a trained image classifier generalises or why its full network makes a decision.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Neural network experiment defense: the case

02. Follow the deciding steps

03. One toy ReLU neuron: not a probability

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.